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BSBI School of Business and Innovation

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1. INTRODUCTION

Retail businesses generate large volumes of operational data, but the value of this data depends on whether it can be transformed into information that supports decision-making. Business intelligence dashboards can support this process by combining performance measures, visual analysis and interactive exploration in a form that managers can interpret quickly. Data storytelling extends this further by connecting visual evidence to a business question and possible action rather than presenting charts in isolation (Gunklach, Jacob and Michalczyk, 2023). AIS eLibrary

This report analyses the **Rossmann Store Sales** dataset using Microsoft Power BI. The dataset was originally released through Kaggle and contains historical daily sales data for **1,115 Rossmann stores in Germany**, together with variables including sales, customer counts, promotions, store type, assortment, competition distance, school holidays and state holidays (Knauer and Cukierski, 2015). Kaggle The final analytical model used **1,017,209 daily store records covering 1 January 2013 to 31 July 2015**.

The business question guiding the analysis is:

How can Rossmann use historical store performance, customer demand patterns, promotions and competitive proximity to improve sales planning and identify stores requiring management attention?

Power BI was used to prepare and model the data, create calculated measures, analyse patterns and develop three connected interactive dashboard pages. The analysis applies **linear regression, six-week trend forecasting and rule-based store segmentation** to support sales planning and store-level decision-making. The dashboards follow a **What–Why–Action** structure: the first evaluates overall performance, the second investigates factors associated with sales performance, and the third supports planning through forecasting and segmentation. This structure moves the analysis from descriptive information towards interpretable insight and practical management action.

2. Literature Review

Business intelligence (BI) combines data preparation, analysis and visual reporting to support evidence-based decision-making. Its effectiveness depends not only on the technology used but also on the quality and accessibility of the underlying information. Işık, Jones and Sidorova (2013) found that capabilities such as data quality, user access and system integration are important contributors to BI success. This is particularly relevant in retail analytics, where inaccurate or poorly structured data can lead to misleading comparisons between stores, periods or customer-demand conditions.

Data visualisation supports decision-making by presenting patterns, comparisons and trends in a form that is easier to interpret than large tables of raw data. However, dashboard quality depends on more than the number of visual elements included. Hjelle *et al.* (2024) found that the format and completeness of dashboard information can influence decision quality by reducing perceived task complexity and improving information satisfaction. This supports the use of focused KPIs, appropriate chart types, consistent visual design and interactive filters rather than overcrowded dashboards.

Data storytelling extends visualisation by connecting analytical evidence with context and a decision that needs to be made. Gunklach, Jacob and Michalczyk (2023) argue that incorporating storytelling techniques into BI dashboards can improve the communication and effective use of analytical insights. This supports a structured narrative in which descriptive performance, explanatory analysis and decision-focused outputs are connected rather than presented as isolated visualisations.

Predictive analysis is also relevant to retail planning because future demand can vary with factors such as seasonality and promotions. Fildes, Ma and Kolassa (2022) highlight the importance and complexity of forecasting in retail decision-making. This provides a suitable theoretical basis for combining historical analysis with forecasting and segmentation in the later stages of this report.

3. Data Exploration and Preprocessing

The Rossmann data was prepared in Microsoft Power BI using two related tables. The daily sales dataset was loaded as **FactSales**, containing **1,017,209 store-day observations**, while store-level metadata was loaded as **DimStore**, representing **1,115 stores**. The dataset covers the period from **1 January 2013 to 31 July 2015**. The main variables consisted of temporal data (Date), numerical measures (Sales, Customers and CompetitionDistance), categorical fields (StoreType, Assortment, Promo, StateHoliday and SchoolHoliday) and the store identifier (Store). The dataset records daily customer counts rather than individual customer-level information and does not provide detailed geographical attributes; these were therefore treated as limitations of the available data.

Data-quality checks were completed before analysis. No duplicate **Store–Date** combinations or unmatched store identifiers were found. The dataset contained **172,817 closed-store records**, all of which recorded zero sales and zero customers. These observations were retained to preserve the complete historical dataset, but measures representing operating performance were calculated using only records where Open = 1. Missing store metadata was not replaced with assumed values. The validation identified **3 missing competition-distance values**, **354 incomplete competition-opening records**, and **544 missing Promo2 timing or interval records**.

Several transformations were then created to support analysis and improve dashboard usability.

Preparation step	**Analytical purpose**
Converted Date to date format	Enabled chronological analysis and forecasting

Created Year, Month, Year-Month and Day Name	Supported trends, sorting and time filters
Created readable Promo, Open and Holiday labels	Improved dashboard interpretation
Calculated Sales per Customer	Supported comparison of sales relative to customer traffic
Converted CompetitionDistance to kilometres and distance bands	Enabled analysis of competitive proximity
Preserved missing values as blank/Unknown	Avoided introducing unsupported information

The model used a **one-to-many relationship between DimStore[Store] and FactSales[Store]**, allowing store characteristics to filter the daily sales observations without duplicating store-level information.

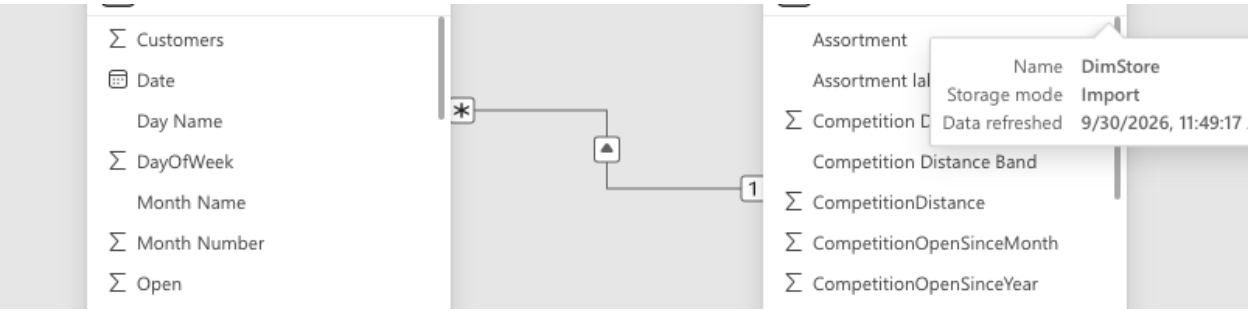


Figure 1. Power BI data model linking store metadata to daily sales observations.

Exploratory validation also identified an important time-series limitation. Only **935 stores were represented between July and December 2014**, meaning comparisons involving this period should be interpreted cautiously because store coverage was not consistent across the full historical period.

4. Advanced Visualisation and Insights

The Power BI report was designed to support both high-level monitoring and interactive investigation. The first page, **Executive Sales Overview**, presents the main operational KPIs for the complete Rossmann dataset: **5.87 billion total sales, 644.04 million customers, average sales of 6,955.51 per open store-day, sales per customer of 9.12, and 1,115 stores**. These measures provide the baseline against which filtered views can be compared.

Interactivity was implemented through slicers for **Year, Store Type and Promo Status**, together with cross-filtering between visuals. Figure 2 demonstrates this by selecting **Store Type A**, causing the KPIs and supporting charts to recalculate automatically.

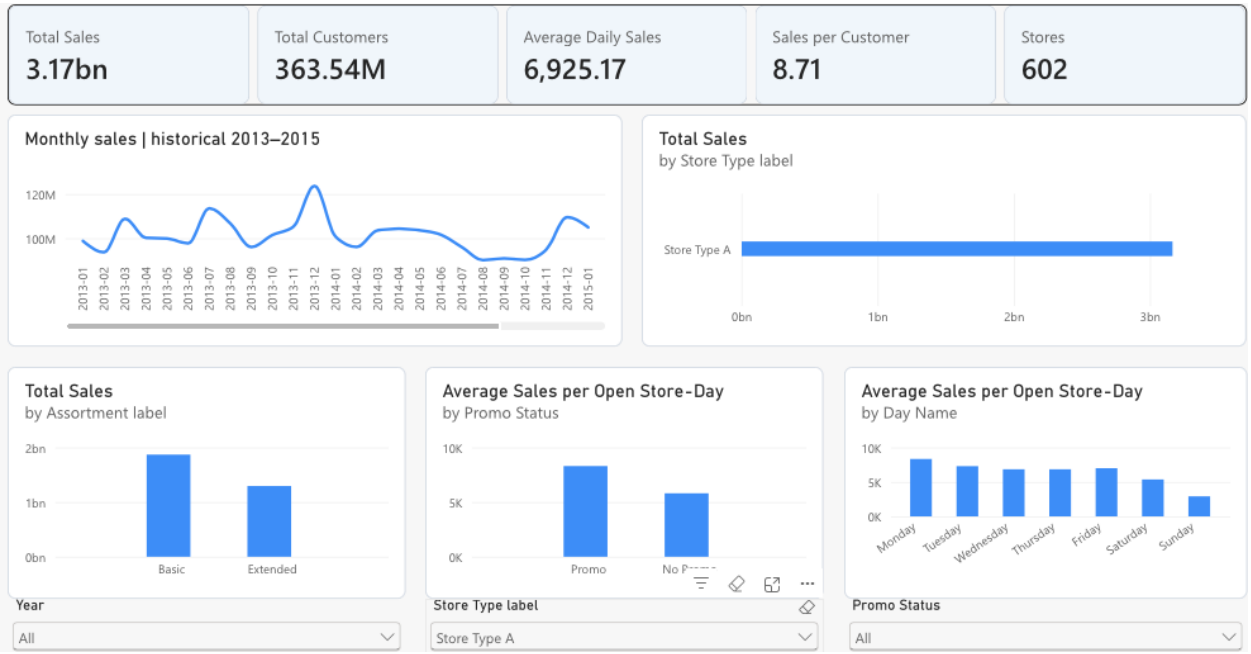


Figure 2. Executive Sales Overview filtered to Store Type A.
Source: Author's Power BI analysis (2026).

Under this filter, the dashboard shows **602 stores, 3.17 billion sales and 363.54 million customers**. Average sales per open store-day were **6,925.17**, while sales per customer were **8.71**. Although Type A stores account for a substantial share of network sales and customer activity, their average operating-day sales remain close to the overall network average and sales per customer are lower than the overall value of 9.12. This suggests that their large contribution is driven partly by scale and customer volume rather than clearly superior sales efficiency.

A combined monthly **Sales and Customers** chart was used as a dual-axis view to compare the movement of both measures over time. This visual supports temporal analysis without requiring separate charts and allows users to identify periods where sales and customer activity move differently. Additional visuals compare sales by

assortment, promotion status and day of week. These provide supporting context rather than separate headline findings.

The second dashboard, **Sales Drivers and Promotion Analysis**, was designed to investigate factors associated with differences in sales performance. Promotional open days recorded average sales of **8,228.28**, compared with **5,929.41** on non-promotional open days, representing an observed difference of approximately **38.8%**. This result indicates a positive association between promotional periods and higher average sales, but it does not demonstrate that promotions alone caused the increase because other factors, including seasonality, store characteristics and customer demand, may also contribute.

Advanced visual techniques were also used to support comparison. A **Store Type × Assortment heat map** highlights stronger and weaker store configurations through conditional formatting, while competition-distance and holiday visuals provide additional contextual views. Together with the slicers and cross-highlighting, these features allow users to move from overall performance to more focused store-level investigation without creating separate reports for each business question.

5. Predictive and Exploratory Analytics

Predictive and exploratory techniques were applied to examine the relationship between customer traffic and sales, identify meaningful store segments and support short-term planning.

5.1 Customer–Sales Regression

A store-level linear regression examined the relationship between **average customers per open store-day** and **average sales per open store-day**.



Figure 3. Store-level customer–sales relationship for Store Type A.
Source: Author's Power BI analysis (2026).

For Store Type A, the regression produced an **R^2 of 0.84** and a **slope of 6.88**. This means that approximately **84% of the variation in average sales per open store-day across Type A stores is accounted for by the fitted linear relationship with average customers per open store-day**.

Store Type	Regression Slope	R^2
A	6.88	0.84
B	4.69	0.77
C	6.85	0.81
D	9.18	0.67

Store Type D produced the steepest slope (**9.18**) but the lowest R^2 (**0.67**). This shows that a stronger fitted increase in sales does not necessarily indicate a more consistent relationship. The regression therefore demonstrates association rather than proving that additional customers directly cause higher sales.

5.2 Store Segmentation

Stores were segmented using network-average thresholds for **customers per open store-day** and **sales per open store-day**. Sales per customer was then used as an additional diagnostic measure to compare the resulting groups.

Segment	Avg Sales / Open Day	Avg Customers	Sales / Customer
High Demand / High Sales	9,500.62	1,136.81	8.36
High Demand / Lower Sales	**6,278.26**	**885.03**	**7.09**
Lower Demand / High Sales	7,756.19	668.34	11.61
Lower Demand / Lower Sales	5,293.72	549.49	9.63

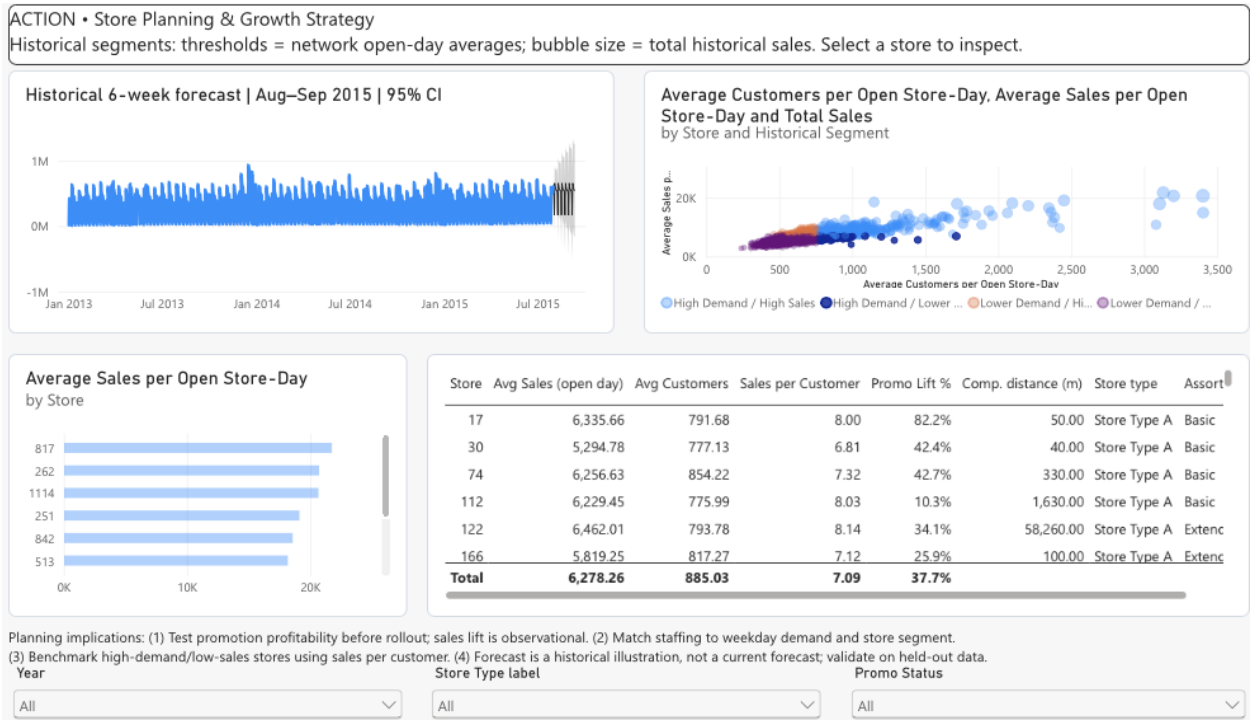


Figure 4. Store Planning and Growth Strategy filtered to the High Demand / Lower Sales segment.
Source: Author's Power BI analysis (2026).

The **High Demand / Lower Sales** group is particularly relevant for management attention. These stores attract relatively high customer traffic but record the **lowest sales per customer at 7.09**. This suggests that the main opportunity may lie less in generating additional traffic and more in improving the value generated from existing demand, for example through assortment, promotional effectiveness or store-level execution.

In contrast, the **Lower Demand / High Sales** segment records the highest sales per customer at **11.61**, despite lower customer traffic. This indicates that some stores generate comparatively stronger value from a smaller customer base.

5.3 Six-Week Sales Forecast

A six-week historical forecast was created for **1 August to 11 September 2015** using Power BI forecasting with weekly seasonality and a **95% confidence interval**.

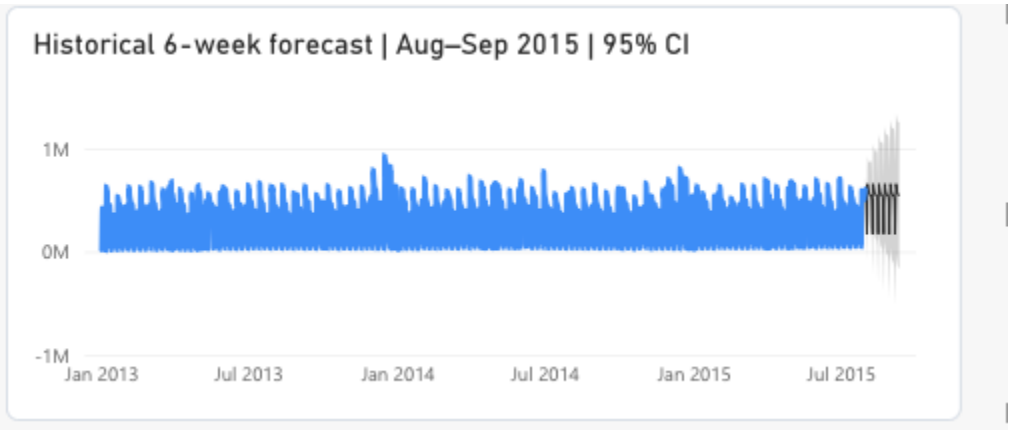


Figure 5. Six-week historical sales forecast with 95% confidence interval.
Source: Author's Power BI analysis (2026).

The forecast provides a short-term planning estimate rather than a guaranteed future outcome. The confidence interval communicates uncertainty around expected sales and can support staffing, promotion and inventory planning. However, because the forecast was not validated against a held-out set of actual observations, it should be interpreted as an indicative planning tool rather than a confirmed measure of predictive accuracy.

6. Dashboard and Story Creation.

The three Power BI pages were designed as a connected **What–Why–Action** story. The **Executive Sales Overview**provides the *What* by summarising sales, customers, store

performance and monthly trends. The **Sales Drivers and Promotion Analysis** provides the *Why* by examining promotional performance, customer–sales relationships and store characteristics. The **Store Planning and Growth Strategy** provides the *Action* through segmentation and short-term forecasting. This structure reflects the principle that data storytelling should connect analytical evidence with a decision context rather than presenting isolated charts (Gunklach, Jacob and Michalczyk, 2023).

The dashboards are interactive rather than static. Slicers for **Year, Store Type and Promo Status** are available across the report, while cross-filtering allows selections in one visual to update related KPIs and charts. Calculated fields including **Sales per Customer, Average Sales per Open Store-Day and Promo Lift** support comparisons between store groups and operating conditions. These features allow managers to move from an overall view to a specific store type, period or promotional status without leaving the report.

The data story can be summarised in five points:

1. **Store Type A contributes a large share of total sales and customer activity, but its sales per customer remain below the network average.**
2. **Promotional open days are associated with 38.8% higher average sales than non-promotional days.**
3. **The customer–sales relationship differs by store type; Store Type A shows a strong fitted relationship ($R^2 = 0.84$), while Store Type D has a steeper slope but weaker fit.**
4. **High Demand / Lower Sales stores represent a management priority because they attract relatively high customer traffic but record the lowest sales per customer at 7.09.**
5. **The six-week forecast provides an indicative planning range for staffing, inventory and promotional decisions while communicating uncertainty through a 95% confidence interval.**

Together, the dashboards move from monitoring performance to investigating possible drivers and identifying areas for management attention. They therefore support the business question by showing **what is happening, why performance differs and where action may be required**.

7. Conclusion and Limitations

This analysis used Power BI to examine historical Rossmann store performance and support sales-planning decisions. The results show that store performance should not be judged from total sales alone. Store Type A contributes a large share of network activity, promotional open days are associated with 38.8% higher average sales, and regression analysis shows that the relationship between customer traffic and sales differs across store types. Segmentation further identified High Demand / Lower Sales stores as a useful management priority because they attract relatively high customer traffic while recording the lowest sales per customer at 7.09.

The analysis therefore suggests that management attention should focus not only on attracting customers, but also on how effectively existing customer demand is converted into sales. The six-week forecast can additionally support short-term staffing, inventory and promotional planning, although its uncertainty should be considered.

Several limitations affect interpretation. The dataset covers 2013–2015, so the findings should not be treated as representative of current Rossmann performance. Store coverage also falls to 935 stores during part of 2014, and some competition and Promo2 metadata are incomplete. The dataset contains aggregate daily customer counts rather than individual customer records and provides limited geographical information. Finally, the regression demonstrates association rather than causation, the segmentation is rule-based rather than statistical clustering, and the forecast was not validated against held-out actual data.

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APPENDIX (if necessary)

Appendix A: Interactive Power BI Report

The complete interactive Power BI report developed for this analysis can be accessed using the link below. The report contains the three dashboard pages: **Executive Sales Overview, Sales Drivers and Promotion Analysis, and Store Planning and Growth Strategy**.

Interactive Power BI report:

https://app.powerbi.com/links/YN4mUf25Fm?ctid=ff28a236-881c-4b29-9b77-71405b66aca2&pbi_source=linkShare

Access may require sign-in using an authorised organisational Microsoft account.